



An introduction to data science

Data science, business, and systems theory

Business and technology

What is a business?

A **business entity** is an **organisation** that uses economic resources to **provide** goods or services to customers in exchange for **money** or other **goods and services**.

Technology in business

Businesses across various sectors are rapidly **adopting technology** to **transform** their operations. This shift has accelerated **development cycles** and given rise to **disruptive business models** and heightened **competition**, highlighting the **essential role** of **technology and automation** in today's business environment.

An overview of the legal aspects

Most people use the terms **business** and **company** interchangeably. Legally, the word **business** describes an **activity** or **project, not a legal entity**. For example, Meta is not a business; it is a company, and its business is social media services.

When a person starts their **own** business, **everything** that it is **belongs** to them. We say that the owner is **liable** for **all** things for which the business is responsible.

Naming a business **doesn't separate** it from its owner, so many businesses **register** as legal **entities**, aligned with corporate laws and regulations that **dictate how** it should be run.

Types of business entities

Sole Proprietorship

A **common** business entity because it is **easy** and **affordable** to set up. A license grants the owner **complete control** over the business, meaning no other person can make decisions for the business and its assets.

It grants the owner **unlimited liability**, which means the owner puts all of their personal assets at risk for the sake of the business.

The owner **can hire** employees that may assist in the running of the business.

Partnership

Also **affordable** though **less easy** to set up. Here, a license means the law recognises **two or more** individuals as the **owner**.

Partners must decide their **share** of the business. The license grants **each owner unlimited liability**, meaning that each puts their assets at risk for the entire business and not only their share.

Types of business entities

Company

While proprietorships and partnerships are tied to their owners, **companies** function as **distinct legal entities**. This means the company itself, rather than its individuals, possesses **ownership** of the business.

This structure provides individuals with a degree of **financial safeguarding**, as they are accountable for only a predetermined portion of the company's debt – a concept known as **limited liability**.

A **private** or **proprietary** company is owned by **one or more shareholders**. A **public** company is the same, but makes its shares available for purchase on a **stock market**.

An overview of protective legislation

Legal requirements are there primarily to **protect people**. The legal entities offer their owners **protection** in some way, but **legislation** also protects **employees**.

Health and safety regulations play a crucial role in ensuring that work environments are **free** from potential **harm** and **risks**, safeguarding employees' well-being.

There are also several **regulations** and **laws** in place to address working hours, leave, contracts, etc., as well as promote **equal opportunities** and **treatment**.

Developing a good business strategy and plan

All organisations need to develop a **strategy** to guide their work. This describes the **overarching goals** of a business and the **actions** upper management will take on behalf of the owners to **meet the goals**.

With strategy as a guide, we can formulate a **business plan**, which is a **working document** that describes **how** the business plans to **implement** its **strategy**.

Steps of a good business plan



An overview of financial concepts

With a business plan in place, the next concept would be **finance**.

Financial statements are **formal documents** that include the **balance sheet**, **income statement**, and **cash flow statement**.

These documents record **how much money** a business has, **where it goes**, **where it comes from**, and **who is using it**.

The **balance sheet** keeps **track** of how much **money** a business has, and what **form** it takes.

The **income statement** (profit and loss reports) details **income** and **expenses** over **regular** periods.

The **cash flow statement** describes **how** the company **uses money** to make **more money** in ways not necessarily directly associated with its chosen business.

An overview of marketing

Marketing consists of all processes involved in **understanding** and **managing** the **exchange** of goods and **services** between **customers** and **providers**.

Most marketing work involves **understanding what consumers want** and **providing business** to them, or **finding** a group of consumers that want your business.



Successful marketing requires a deep **alignment** between **consumer needs** and **business offerings**.

An introduction to technology

Technology has always been vitally important to **successful businesses**, especially those technologies aimed at improving **efficiency**.

Any business that can **access** and **share information faster** than its **competitors** will have a significant **advantage**.

The utilisation of **information systems** has become important for **efficient operations**. These systems span a spectrum of functionalities, each catering to **specific organisational needs** and **levels**.

Preparing for the meeting

For a typical organisation, there are usually **six** different types of **information systems**.

1 Transaction Processing Systems (TPS):
Process routine and operational transactions, such as sales and inventory updates, at the operational level of an organisation.

3 Knowledge Management Systems (KMS):
Facilitate the collection, organisation, and dissemination of knowledge, fostering better decision-making and collaboration.

5 Decision Support Systems (DSS): assist managers in making semi-structured and unstructured decisions by providing tools and access to data and models.

2 Office Automation Systems (OAS): Streamline and automate everyday office tasks, such as document management, communication, and scheduling, to enhance office efficiency.

4 Management Information Systems (MIS):
Gather, analyse, and present summarised information to middle managers, aiding in tactical decision-making and planning.

6 Executive Support Systems (ESS): provide insights to senior executives to support strategic decisions, focusing on trends and external factors impacting the organisation.

How to provide and respond to feedback

The ability to **compile, provide, and respond** to feedback is vital when it comes to working with other people in a team, family, or community – **regardless** of the **objective** of that group.

In order to work effectively, people need to **know, understand, and acknowledge** how their behaviours and attitudes **impact** those around them.

Feedback considerations

When providing feedback to people, the feedback provider should be aware of the **context**, **situation**, and **details** of the actions that are being commented on. The following should be taken into account:

Focus on a particular behaviour

This should be **specific** rather than general.

Minimise blame statements

Focusing on the **impact of an action** rather than blame can avoid emotional responses.

Not character associated

Feedback should be **behaviour or action-related**, not a criticism of someone's personality.

Ensure understanding

The provider should make sure that the recipient **understands the feedback** properly.

Outcome orientated

Focus on the **details of what happened**, i.e. awareness of the outcome or impact is important.

Well timed

Feedback is most effective when there is a **short time between behaviour and feedback**.

Appropriate work behaviour

All places of work have a **different work culture** – a different atmosphere. This can be passed down from management in terms of **company values** as well as the company's **code of conduct**.

All companies have a **code of conduct** which defines the employees' **day-to-day behaviour**. Think of the ExploreAI honour code as our **code of conduct**.

This reflects our **culture, values**, and **what is expected** of us.

Appropriate work behaviour guidelines

- Put in the **effort**.
- Be **present**.
- Use the **resources** made available to us.
- **Share** our **knowledge** for the benefit of our fellow students.
- **Communicate** openly and honestly with our teammates and supervisor.
- Stick to **deadlines**.
- Take our **teamwork** seriously.